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Explain how HIV affects the production of antibodies when AIDS develops in a person.
[3 marks]

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A scientist measured the effect of a drug on the number of T cells and the number of HIV particles in blood taken from a person with AIDS. The results are shown in **Figure 5**.

The graph displays two data series over a 16-month period. The left y-axis represents the number of T cells per mm³ of blood, ranging from 0 to 600. The right y-axis represents the number of HIV particles per mm³ of blood, ranging from 0 to 120. The x-axis represents time in months, from 0 to 16. The T cell count (solid line) starts at approximately 40, rises to a peak of about 110 at 6 months, and then fluctuates between 80 and 120. The HIV particle count (dashed line) starts at approximately 60, peaks at about 75 at 2 months, drops sharply to about 35 at 4 months, and then steadily increases back to about 85 by 16 months.

Time since start of drug treatment / months	Number of T cells per mm ³ of blood	Number of HIV particles per mm ³ of blood
0	40	60
2	100	75
4	80	35
6	110	55
8	100	70
10	90	80
12	80	85
14	80	90
16	120	85

--- T cells
— HIV particles

Use all of this information to evaluate the effectiveness of the drug in treating AIDS.

[5 marks]

[illegible]

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